

Autogene

Autogene is a synthetic-biology / systems-biology helper that turns gene names or DNA sequences into an interactive regulatory and interaction graph centered on *E. coli* knowledge from a local EcoCyc PGDB. Users supply seeds (e.g. *lacI*, *lacZ*) or FASTA/raw DNA; the tool resolves genes against EcoCyc when possible, falls back to UniProt for unknown symbols, and augments the graph with protein–protein interactions from UniProt and STRING. A Flask backend (`circuit_viewer/server.py`) exposes APIs and serves a static circuit viewer UI. Optional BLASTp (local or NCBI web) suggests CDS-associated gene symbols from imported sequences. The project is Python-based and is run with the interpreter rather than a traditional compile step.

Major software components

Function	Usage
<code>ecocyc_client.py</code>	Loads a local EcoCyc flat-file PGDB (<code>genes.dat</code> , <code>proteins.dat</code> , <code>regulation.dat</code> , etc.), indexes names and regulation, and resolves user tokens to EcoCyc frame IDs (with UniProt fallback via <code>remote_clients</code>).
<code>graph_builder.py</code>	Builds a NetworkX directed graph from resolved seeds: EcoCyc regulation edges plus UniProt / STRING interaction partners; can trim paths between seeds and export JSON for the viewer.
<code>remote_clients.py</code>	HTTP clients for UniProt REST and STRING APIs: search best hit, fetch entry JSON, and map accession/symbol to stable graph node IDs and URLs.
<code>cli.py</code>	Command-line entry: resolve gene tokens, build the graph, print a summary, and write <code>circuit_data.json</code> for offline use with the viewer.
<code>circuit_viewer/server.py</code>	Flask + CORS server: imports the facade from <code>Autogene.py</code> , serves the web UI, and exposes JSON APIs (including sequence import hints).

Special Instructions

Users would need to install with pip, e.g. `pip install NetworkX flask flask-cors`; add `biopython` in order to use BLASTP in order to find the protein. In addition to that, users would need to request for access to the different databases (Ecocyc and UNIPROT) themselves, either by applying for an educational license through their academic institution, or paying a fee. This project primarily runs locally without a need to access the internet, but if a user wants to use the fallback feature to search for the gene/protein online in the case that they are not found in the local databases, then they would need the internet for that.